

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : PRINCIPLES OF OPERATING SYSTEM

Practical No : 9

Aim : Memory Management Techniques

- Simulate FIFO and LRU page replacement using page reference strings.
- Measure hit/miss ratios under different reference patterns.
- Extend to include frames and memory constraints.

CODE :

```
from collections import deque
```

```
print("Om Wala S119")
print("MEMORY MANAGEMENT TECHNIQUES")
print("-" * 50)
```

```
def fifo_page_replacement(pages, frames):
```

```
    memory = []
```

```
    queue = deque()
```

```
    hits = 0
```

```
    misses = 0
```

```
    for page in pages:
```

```
        if page in memory:
```

```
            hits += 1
```

```
        else:
```

```
            misses += 1
```

```
        if len(memory) < frames:
```

```
            memory.append(page)
```

```
            queue.append(page)
```

```
        else:
```

```
            old_page = queue.popleft()
```

```
            memory.remove(old_page)
```

```
            memory.append(page)
```

```
            queue.append(page)
```

```
    hit_ratio = hits / len(pages)
```

```
    miss_ratio = misses / len(pages)
```

```
    return hits, misses, hit_ratio, miss_ratio
```

```
def lru_page_replacement(pages, frames):
```

```
    memory = []
```

```
    hits = 0
```

```
    misses = 0
```

```
    for page in pages:
```

```
        if page in memory:
```

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```
        hits += 1
        memory.remove(page)
        memory.append(page)
    else:
        misses += 1

    if len(memory) >= frames:
        memory.pop(0)

    memory.append(page)

    hit_ratio = hits / len(pages)
    miss_ratio = misses / len(pages)

    return hits, misses, hit_ratio, miss_ratio

def display_results(pages, frames):
    fifo_hits, fifo_misses, fifo_hit_ratio, fifo_miss_ratio = \
        fifo_page_replacement(pages, frames)

    lru_hits, lru_misses, lru_hit_ratio, lru_miss_ratio = \
        lru_page_replacement(pages, frames)

    print("\nPage Reference String:")
    print(pages)

    print("\nNumber of Frames:", frames)

    print("\nFIFO Page Replacement")
    print("Hits      :", fifo_hits)
    print("Misses     :", fifo_misses)
    print("Hit Ratio  :", round(fifo_hit_ratio, 2))
    print("Miss Ratio :", round(fifo_miss_ratio, 2))

    print("\nLRU Page Replacement")
    print("Hits      :", lru_hits)
    print("Misses     :", lru_misses)
    print("Hit Ratio  :", round(lru_hit_ratio, 2))
    print("Miss Ratio :", round(lru_miss_ratio, 2))

pages = list(map(int, input(
    "\nEnter page reference string (space separated): "
).split()))

frames = int(input("Enter number of memory frames: "))
```

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```
if frames <= 0:
    print("Number of frames must be greater than 0.")
else:
    display_results(pages, frames)

    print("\n" + "-" * 50)
    print("COMPARISON FOR DIFFERENT MEMORY FRAME SIZES")
    print("-" * 50)

    print("\nFrames | FIFO Hits | FIFO Misses | LRU Hits | LRU Misses")
    print("-" * 55)

    for f in range(1, frames + 1):
        fh, fm, _, _ = fifo_page_replacement(pages, f)
        lh, lm, _, _ = lru_page_replacement(pages, f)

        print(f"{f:^6} | {fh:^9} | {fm:^11} | {lh:^8} | {lm:^10}")
```

OUTPUT :

```
===== RESTART: C:/Python/OS/pract9.py =====
Om Wala S119
MEMORY MANAGEMENT TECHNIQUES
-----

Enter page reference string (space separated): 1 2 3 4 2 1 3 5 1
Enter number of memory frames: 3

Page Reference String:
[1, 2, 3, 4, 2, 1, 3, 5, 1]

Number of Frames: 3

FIFO Page Replacement
Hits      : 3
Misses    : 6
Hit Ratio : 0.33
Miss Ratio: 0.67

LRU Page Replacement
Hits      : 2
Misses    : 7
Hit Ratio : 0.22
Miss Ratio: 0.78

-----
COMPARISON FOR DIFFERENT MEMORY FRAME SIZES
-----

Frames | FIFO Hits | FIFO Misses | LRU Hits | LRU Misses
-----
1      | 0         | 9           | 0         | 9
2      | 0         | 9           | 0         | 9
3      | 3         | 6           | 2         | 7
>>>|
```

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